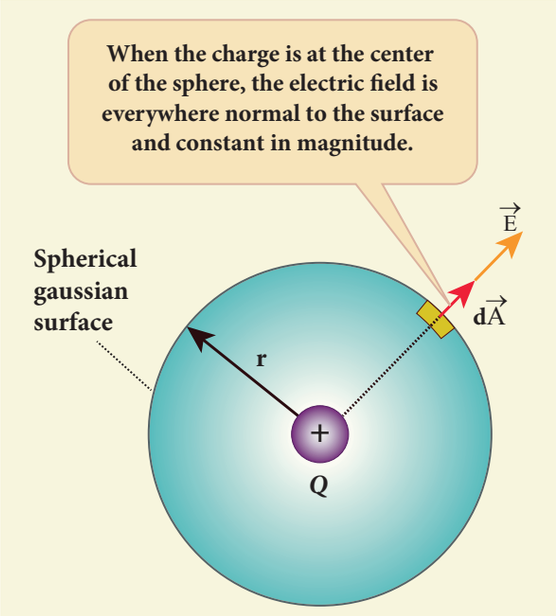


Figure 1.34 Total electric flux of point charge

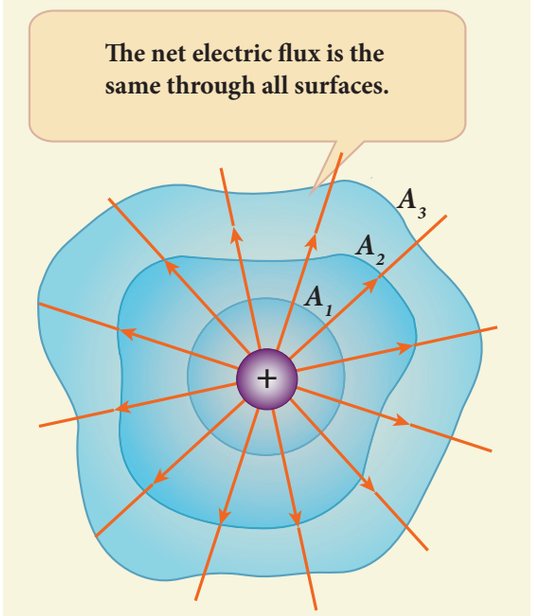


Figure 1.35 Gauss law for arbitrarily shaped surface

E is uniform on the surface of the sphere,

$$\Phi_E = E \oint dA \qquad (1.60)$$

Substituting for $\oint dA = 4\pi r^2$ and $E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$ in equation (1.60), we get

$$\Phi_E = \frac{1}{4\pi\epsilon_\circ}\frac{Q}{r^2} \times 4\pi r^2 = 4\pi \frac{1}{4\pi\epsilon_\circ} Q$$

$$\Phi_E = \frac{Q}{\epsilon_\circ} \qquad (1.61)$$

The equation (1.61) is called as Gauss's law.

The remarkable point about this result is that the equation (1.61) is equally true for any arbitrary shaped surface which encloses the charge Q and as shown in the Figure 1.35. It is seen that the total electric flux is the same for closed surfaces A_1, A_2 and A_3 as shown in the Figure 1.35.

Gauss's law states that **if a charge Q is enclosed by an arbitrary closed surface, then the total electric flux Φ_E through the closed surface is**

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{encl}}{\epsilon_\circ} \qquad (1.62)$$

where Q_{encl} denotes the charges within the closed surface.

Discussion of Gauss law

(i) The total electric flux through the closed surface depends only on the charges enclosed by the surface and the charges present outside the surface will not contribute to the flux and the shape of the closed surface which can be chosen arbitrarily.

(ii) The total electric flux is independent of the location of the charges inside the closed surface.

12th Physics_Vol-1_Unit 1_English Medium.indd 40 08-12-2022 11:03:38